DSE Conformance Report

Velexa matching engine vs the order types the OMS sends to DSE · generated 17 October 2026

Scope. 184 cases were driven through the engine's real FIX acceptor (FIXT.1.1 / FIX 5.0 SP2, strict data dictionary on) using the exact tag set listed in DSE FIX Order Types · Velexa OMS (11 Oct 2026). Expected DSE behaviour comes from that document, its observed DSE rejects (price band ±7.5% / ±8.75%, minimum quantity step 1000) and the FIX specification. Where DSE behaviour is not specified the case is marked Info and not judged. Findings are classified from this run's results, so a fixed problem appears under Resolved. Reproduce with `go test -run TestConformance` then `python3 tools/conformance_report.py`.

184	146	0	26	12
cases run	match DSE	deviate (blockers)	gaps	partial / info

Open findings

Finding	Status	Detail
Validity At-Open / Good-till-Crossing / At-Close TIF-2, TIF-5, TIF-7	Gap	59=2, 5 and 7 are accepted but treated as Day.
ExecInst (18): Park / Unpark / Block EI-1 … EI-6	Gap	Tag 18 is not interpreted, and the DSE combinations 'w S' and 'w q' are not in the data dictionary.
MinQty (110) QO-3, QO-4	Gap	MinQty is ignored: an order can fill below it, and MinQty > OrderQty is accepted.
Board groups (762) BG-02 … BG-17	Gap	The 16 non-PUBLIC groups are accepted on a MAIN-board equity (including INDEX, ATBX, SCPX, debt and block groups); tag 762 is not validated.
Automatic circuit breaker; category and board rules ST-3, ST-4	Gap	No automatic volatility halt; no category- or board-specific lot, band or session rules.

Resolved

Finding	Status	Now
Orders that DSE would refuse are accepted OT-02, OT-095, OT-099, SD-4	Matches DSE	Limit orders need a price; unsupported OrdTypes and Sides are rejected with a reason.
Fill reports and cancel rejects are valid FIX ER-2, CM-3	Matches DSE	Fills carry 150=F; 35=9 carries OrderID (the real one when the order exists).

Finding	Status	Now
Order Status Request (35=H) CM-11, CM-12	Matches DSE	Answered with 150=I and the current status; unknown orders get 39=8.
Amend (35=G) pre-trade checks CM-7, CM-8	Matches DSE	Amends are checked like new orders (tick, band, lot, session, ClOrdID) and refused with 35=9 and the reason.
Duplicate ClOrdID QT-6	Matches DSE	Rejected while the first order is working; ClOrdIDs are tracked per broker.
Pre-Open auction PH-3, PH-4	Matches DSE	Price checks apply in Pre-Open; auction fills are reported to both brokers.
GTD and Day expiry; post-close session TIF-6a, TIF-6b, ST-1, ST-2	Matches DSE	GTD needs a future expiry; expiry sends 150=C; Day orders expire at the end of the day; post-close trades at the close price only.
Market-At-Best, Validity S, DisplayQty OT-06, TIF-S, QO-1	Matches DSE	Z trades at the best opposite price; S is treated as Day; DisplayQty sets the iceberg peak.

What works: limit, market and market-at-best matching with price-time priority, partial fills, IOC, FOK and GTD, cancel, amend and status of working orders, the full DSE price-band ladder in every session and on amends, tick-size enforcement, halted and unknown symbols, whole-share quantities, the per-instrument market lot, duplicate-ClOrdID protection, required-field validation, the Pre-Open auction, day-order expiry and the post-close session, and the session messages Logon / Logout / TestRequest / ResendRequest / SequenceReset.

Results by section

Section	Cases	Match	Deviate	Gap	Partial / info
Order type (40)	10	7	0	0	3
Validity (59)	14	11	0	3	0
Side & quantity	13	13	0	0	0
Board group (762)	20	4	0	16	0
ExecInst (18)	6	0	0	3	3
Quantity options	4	1	0	2	1
Tags & reports	14	13	0	0	1
Price band / tick	70	69	0	0	1
Cancel / modify / status	13	12	0	0	1
Phases	5	5	0	0	0
Matching	3	3	0	0	0
Rules not modelled	6	2	0	2	2
Session messages	6	6	0	0	0

Comparison: OMS / DSE vs engine

1. Order type (40)

ID	Case	DSE / OMS expects	Matching engine did	Result
OT-01	Limit (40=2) with price	Accepted, rests on book	`35=8 150=0 39=0`	Matches DSE
OT-02	Limit (40=2) without price (44), liquid book	Rejected: price required for a limit order	`35=8 150=8 39=8 58=limit order without a price (tag 44)`	Matches DSE
OT-03	Market (40=1) without price, liquid book	Matches at best available price	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
OT-04	Market (40=1) carrying a price outside the post-close session	Rejected: 'Price is not allowed for market order' (a price is only allowed, and must be the close, in post-close)	`35=8 150=8 39=8 58=Price is not allowed for market order`	Matches DSE
OT-05	Market (40=1) on an empty book	Order not executed (no liquidity); treatment of the remainder not specified by the OMS reference	`35=8 150=8 39=8 58=no liquidity: no resting SELL orders on T004M` rejected with 'no liquidity' — market orders never rest	Info
OT-06	Market-At-Best (40=Z)	DSE-specific type: accepted and matched at best	`35=8 150=0 39=0 ; 35=8 150=F 39=1 14=60`	Matches DSE
OT-07	Stop (40=3) with StopPx (99)	Not used by the OMS (informational)	`35=8 150=A 39=A` engine: 35=8 150=A 39=A	Info
OT-08	Stop-limit (40=4) with StopPx and price	Not used by the OMS (informational)	`35=8 150=A 39=A` engine: 35=8 150=A 39=A	Info
OT-095	Unsupported OrdType 40=5	Rejected as an unsupported order type	`35=8 150=8 39=8 58=unsupported order type (tag 40): "5"`	Matches DSE
OT-099	Unsupported OrdType 40=9	Rejected as an unsupported order type	`35=8 150=8 39=8 58=unsupported order type (tag 40): "9"`	Matches DSE

2. Validity (59)

ID	Case	DSE / OMS expects	Matching engine did	Result
TIF-0	Day (59=0) resting limit	Accepted, rests until close	`35=8 150=0 39=0`	Matches DSE
TIF-1	GTC (59=1) resting limit	Accepted, rests until cancelled	`35=8 150=0 39=0`	Matches DSE

ID	Case	DSE / OMS expects	Matching engine did	Result
TIF-3a	IOC (59=3) partially fillable (50 of 100)	Fills 50, cancels the remaining 50	`35=8 150=0 39=0 ; 35=8 150=F 39=1 14=50 ; 35=8 150=4 39=4 14=50`	Matches DSE
TIF-3b	IOC (59=3) with no liquidity	Cancelled / expired with no fill	`35=8 150=0 39=0 ; 35=8 150=4 39=4`	Matches DSE
TIF-4a	FOK (59=4) fully fillable	Fully filled	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
TIF-4b	FOK (59=4) insufficient liquidity	Whole order cancelled, no partial fill	`35=8 150=0 39=0 ; 35=8 150=4 39=4`	Matches DSE
TIF-6a	GTD (59=6) without expiry date (OMS sends none)	Rejected without ExpireDate(432)/ ExpireTime(126) — OMS reference says DSE 'will probably reject'	`35=8 150=8 39=8 58=GTD order without an expiry: send ExpireDate (432) or ExpireTime (126)`	Matches DSE
TIF-6b	GTD (59=6) with ExpireTime (126) 3 s ahead — expires	Order removed at expiry AND an Execution Report (150=C Expired) is sent	`35=8 150=0 39=0 ; 35=8 150=C 39=C`	Matches DSE
TIF-6c	GTD (59=6) with ExpireDate (432) only	Accepted; expires at end of that date	`35=8 150=0 39=0`	Matches DSE
TIF-2	At the Opening (59=2)	Honoured as its own validity (not offered by any OMS screen, but the builder can pass it)	`35=8 150=0 39=0` accepted but silently treated as Day (mapTIF default) — semantics not implemented	Gap
TIF-5	Good Till Crossing (59=5)	Honoured as its own validity (not offered by any OMS screen, but the builder can pass it)	`35=8 150=0 39=0` accepted but silently treated as Day (mapTIF default) — semantics not implemented	Gap
TIF-7	At the Close (59=7)	Honoured as its own validity (not offered by any OMS screen, but the builder can pass it)	`35=8 150=0 39=0` accepted but silently treated as Day (mapTIF default) — semantics not implemented	Gap
TIF-S	Session (59=S) — offered by the OMS trader UI	Accepted (the OMS offers it); not a FIX 5.0 SP1/SP2 enum value	`35=8 150=0 39=0`	Matches DSE
TIF-X	Missing TimeInForce	Defaults to Day	`35=8 150=0 39=0`	Matches DSE

3. Side & quantity

ID	Case	DSE / OMS expects	Matching engine did	Result
SD-1	Buy (54=1)	Accepted	`35=8 150=0 39=0`	Matches DSE
SD-2	Sell (54=2)	Accepted	`35=8 150=0 39=0`	Matches DSE
SD-3	Missing Side (54)	Rejected	`35=3 371=54 372=D 373=1 58=Required tag missing`	Matches DSE

ID	Case	DSE / OMS expects	Matching engine did	Result
SD-4	Side 54=5 (sell short)	Rejected (the OMS reference: no short-sell side)	`35=8 150=8 39=8 58=unsupported side (tag 54): "5" — only 1 (buy) and 2 (sell) are traded`	Matches DSE
QT-1	OrderQty = 0	Rejected	`35=8 150=8 39=8 58=invalid OrderQty (tag 38): 0 — quantity must be greater than zero`	Matches DSE
QT-2	Fractional OrderQty (10.5)	Rejected (whole shares only)	`35=8 150=8 39=8 58=invalid OrderQty (tag 38): 10.5 — the exchange trades whole shares only`	Matches DSE
QT-3	Negative OrderQty	Rejected	`35=8 150=8 39=8 58=invalid OrderQty (tag 38): -5 — quantity must be greater than zero`	Matches DSE
QT-4	Quantity a multiple of 1000	Accepted	`35=8 150=0 39=0`	Matches DSE
QT-5	Quantity 1500 on a lot-1000 instrument	Rejected: '(792) Minimum quantity step: 1000' (observed DSE reject, instrument-dependent)	`35=8 150=8 39=8 58=(792): Minimum quantity step: 1000`	Matches DSE
QT-5b	Quantity 2000 on a lot-1000 instrument	Accepted (whole lots)	`35=8 150=0 39=0`	Matches DSE
QT-6	Duplicate ClOrdID (11) resubmitted	Rejected as a duplicate order	`35=8 150=8 39=8 58=Duplicate ClOrdID: already used by a working order: CF00035`	Matches DSE
QT-7	Missing Symbol (55)	Rejected	`35=8 150=8 39=8 58=missing Symbol (tag 55)`	Matches DSE
QT-8	Unknown symbol	Rejected (not in the security master)	`35=8 150=8 39=8 58=unknown symbol NOSUCHSYM: not in this engine's instrument universe (unknown)`	Matches DSE

4. Board group (762)

ID	Case	DSE / OMS expects	Matching engine did	Result
BG-01	762=PUBLIC on a MAIN-board equity	Accepted (public market, MAIN board)	`35=8 150=0 39=0`	Matches DSE
BG-02	762=ATBPUB on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-03	762=SPUBLIC on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-04	762=BLOCK on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap

ID	Case	DSE / OMS expects	Matching engine did	Result
BG-05	762=SBLOCK on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-06	762=BUYIN on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-07	762=ATBBUYIN on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-08	762=SBUYIN on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-09	762=DEBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-10	762=ATBDEBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-11	762=SDEBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-12	762=ALTDBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-13	762=BUYDBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-14	762=YIELDDBT on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-15	762=INDEX on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-16	762=ATBX on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap

ID	Case	DSE / OMS expects	Matching engine did	Result
BG-17	762=SCPX on a MAIN-board equity	Group must match the instrument's board/ segment — rejected on a MAIN-board equity	`35=8 150=0 39=0` accepted — tag 762 is never read, so board groups are not validated against the instrument	Gap
BG-18	762=SPUBLIC on an SME-board equity	Accepted	`35=8 150=0 39=0`	Matches DSE
BG-19	762=ATBPUB on an ATB-board equity	Accepted	`35=8 150=0 39=0`	Matches DSE
BG-20	Missing 762	Accepted (defaults to the instrument's group)	`35=8 150=0 39=0`	Matches DSE

5. ExecInst (18)

ID	Case	DSE / OMS expects	Matching engine did	Result
EI-1	18=Placed (n) on a new order	Normal placed order — rests on the book	`35=8 150=0 39=0` accepted; instruction not interpreted (tag 18 never read)	Partial
EI-2	18=Park (S) on a new order	Order parked: held off the book, no matching	`35=8 150=0 39=0` accepted but the order rests on the book — Park is not implemented (tag 18 never read)	Gap
EI-3	18=Block (w) on a new order	Block-board order	`35=8 150=0 39=0` accepted; instruction not interpreted (tag 18 never read)	Partial
EI-4	18=Block Park (w S) on a new order	Block order parked	`35=3 371=18 372=D 373=5 58=Value is incorrect (out of range) for this tag` rejected — 'w S' is not in the engine's tag-18 dictionary enum, so the OMS's DSE-specific instruction never reaches the engine	Gap
EI-5	18=q (Unpark) on a modify (35=G)	Un-parks the order and places it on the book	`35=8 150=5 39=0` accepted as an ordinary amend; instruction ignored	Partial
EI-6	18=w q (Unpark) on a modify (35=G)	Un-parks the order and places it on the book	`35=3 371=18 372=G 373=5 58=Value is incorrect (out of range) for this tag` rejected — Unpark ('w q') cannot be sent through the dictionary-validated session	Gap

6. Quantity options

ID	Case	DSE / OMS expects	Matching engine did	Result
QO-1	DisplayQty (1138=30) iceberg on a 100-lot (the OMS's tag)	Only 30 shown on the book	`35=8 150=0 39=0`	Matches DSE
QO-2	MaxFloor (111=30) iceberg (engine's own tag)	(not sent by the OMS) 30 shown	`35=8 150=0 39=0` 111 honoured — book shows 30	Info

ID	Case	DSE / OMS expects	Matching engine did	Result
Q0-3	MinQty (110) greater than OrderQty (500 > 100)	Rejected	`35=8 150=0 39=0` accepted — MinQty (110) is ignored	Gap
Q0-4	MinQty (110=50) with only 30 available	No fill (minimum quantity cannot be met)	`35=8 150=0 39=0 ; 35=8 150=F 39=1 14=30` filled below the minimum quantity — MinQty (110) is ignored	Gap

7. Tags & reports

ID	Case	DSE / OMS expects	Matching engine did	Result
TG-1	Party group 453 (448/447/452) present	Accepted	`35=8 150=0 39=0`	Matches DSE
TG-2	No party group	Accepted	`35=8 150=0 39=0`	Matches DSE
TG-3	OrderRestrictions 529=N (non-standard)	Accepted (DSE accepted it in logged traffic)	`35=8 150=0 39=0`	Matches DSE
TG-4	Missing Account (1)	Accepted or rejected per DSE (BO ID required by the OMS)	`35=8 150=0 39=0` engine: 35=8 150=0 39=0	Info
TG-5	Missing TransactTime (60)	Rejected (required field)	`35=3 371=60 372=D 373=1 58=Required tag missing`	Matches DSE
TG-6	Missing ClOrdID (11)	Rejected	`35=3 371=11 372=D 373=1 58=Required tag missing`	Matches DSE
ER-1	Fill ER carries FIX-required tags (37,17,150,39,55,54,151,14,11)	All present	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
ER-2	Fill ER ExecType (150)	150=F (Trade)	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
ER-3	ER echoes Account (1)	Echoed (OMS correlates by BO ID)	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
ER-4	ER echoes OrdType (40) / TimeInForce (59)	Echoed	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
ER-5	ER carries last fill price/qty (31/32)	31 and 32 on a fill	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
ER-6	ER carries average price (6) and cumulative qty (14)	Both present	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
ER-7	ER identifies the instrument by Symbol (55) only, as the DSE does	No 48/22/207/1300 (the DSE sends none)	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
ER-8	Reject ER carries OrdRejReason (103) and Text (58)	103 + 58 on every rejected order	`35=8 150=8 39=8 58=price band: 50.00 outside +/-10.00% limit [90.00-110.00] of prev 100.00`	Matches DSE

8. Price band / tick

ID	Case	DSE / OMS expects	Matching engine did	Result
PT-1	Price on tick (100.10)	Accepted	`35=8 150=0 39=0`	Matches DSE
PT-2	Price off tick (100.05)	Rejected: off-tick	`35=8 150=8 39=8 58=off-tick: 100.0500 is not a multiple of tick 0.10`	Matches DSE
PT-3	Price 100.02 (sub-paisa tick)	Rejected: off-tick	`35=8 150=8 39=8 58=off-tick: 100.0200 is not a multiple of tick 0.10`	Matches DSE
PT-4	Market order ignores the price band	DSE applies circuit limits to execution prices	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100` market orders skip band/tick checks (engine.go:297 applies them to limits only): 35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100	Info
PT-5	Order on a halted instrument	Rejected	`35=8 150=8 39=8 58=trading halted in T046M`	Matches DSE
PT-6	Order after the halt is lifted	Accepted	`35=8 150=0 39=0`	Matches DSE

DSE circuit-limit tiers — boundary test (64/64 correct)

Each tier is probed at the last accepted price and the next tick beyond it, both sides. Tiers: ±10% to Tk200, ±8.75% to Tk500, ±7.5% to Tk1000, ±6.25% to Tk2000, ±5% to Tk5000, ±3.75% above.

Previous close	Band	At upper limit	Just above	At lower limit	Just below
Tk 100.00	±10.00%	110.00 ✔	110.10 ✔	90.00 ✔	89.90 ✔
Tk 200.00	±10.00%	220.00 ✔	220.10 ✔	180.00 ✔	179.90 ✔
Tk 200.10	±8.75%	217.60 ✔	217.70 ✔	182.60 ✔	182.50 ✔
Tk 300.00	±8.75%	326.20 ✔	326.30 ✔	273.80 ✔	273.70 ✔
Tk 500.00	±8.75%	543.70 ✔	543.80 ✔	456.30 ✔	456.20 ✔
Tk 500.10	±7.50%	537.60 ✔	537.70 ✔	462.60 ✔	462.50 ✔
Tk 800.00	±7.50%	860.00 ✔	860.10 ✔	740.00 ✔	739.90 ✔
Tk 1000.00	±7.50%	1075.00 ✔	1075.10 ✔	925.00 ✔	924.90 ✔
Tk 1000.10	±6.25%	1062.60 ✔	1062.70 ✔	937.60 ✔	937.50 ✔
Tk 1500.00	±6.25%	1593.70 ✔	1593.80 ✔	1406.30 ✔	1406.20 ✔
Tk 2000.00	±6.25%	2125.00 ✔	2125.10 ✔	1875.00 ✔	1874.90 ✔
Tk 2000.10	±5.00%	2100.10 ✔	2100.20 ✔	1900.10 ✔	1900.00 ✔
Tk 3000.00	±5.00%	3150.00 ✔	3150.10 ✔	2850.00 ✔	2849.90 ✔
Tk 5000.00	±5.00%	5250.00 ✔	5250.10 ✔	4750.00 ✔	4749.90 ✔
Tk 5000.10	±3.75%	5187.60 ✔	5187.70 ✔	4812.60 ✔	4812.50 ✔
Tk 8000.00	±3.75%	8300.00 ✔	8300.10 ✔	7700.00 ✔	7699.90 ✔

9. Cancel / modify / status

ID	Case	DSE / OMS expects	Matching engine did	Result
CM-1	Cancel a resting order (35=F)	Execution Report 150=4 Cancelled	`35=8 150=4 39=4`	Matches DSE
CM-2	Cancel an unknown order	Order Cancel Reject (35=9)	`35=9 58=(226): Invalid order number 102=99`	Matches DSE
CM-3	35=9 carries OrderID (37) — required by FIX	37 present on the reject	`35=9 58=(226): Invalid order number 102=99`	Matches DSE
CM-4	Modify price WITHOUT Symbol/762 (as the OMS sends 35=G)	Execution Report 150=5 Replaced	`35=8 150=5 39=0`	Matches DSE
CM-5	Modify quantity 100 -> 60	150=5 Replaced with the new quantity	`35=8 150=5 39=0`	Matches DSE
CM-6	Modify an unknown order	35=9	`35=9 58=(226): Invalid order number 102=99`	Matches DSE
CM-7	Modify to a price off tick (96.05)	Rejected: off-tick	`35=9 58=off-tick: 96.0500 is not a multiple of tick 0.10 102=99`	Matches DSE
CM-8	Modify to a price outside the band (150.00)	Rejected: outside price band	`35=9 58=price band: 150.00 outside +/-10.00% limit [90.00-110.00] of prev 100.00 102=99`	Matches DSE
CM-9	Modify using the latest ClOrdID chain (replace a replaced order)	Accepted	`35=8 150=5 39=0`	Matches DSE
CM-10	Replace ER echoes OrigClOrdID (41)	41 echoed on the replace/cancel ER	`35=8 150=5 39=0`	Matches DSE
CM-11	Order Status Request (35=H) on a live order	Execution Report with the current status (150=I)	`35=8 150=I 39=0`	Matches DSE
CM-12	Order Status Request (35=H) on an unknown order	Execution report 150=I, 39=8 (or a reject)	`35=8 150=I 39=8 58=(226): Invalid order number`	Matches DSE
CM-13	Modify a parked stop order	(not used by the OMS) Replaced	`35=9 58=(226): Invalid order number 102=99` engine: 35=9 58=(226): Invalid order number 102=99	Info

10. Phases

ID	Case	DSE / OMS expects	Matching engine did	Result
PH-1	Limit order during Pre-Open	Accepted into the call auction, no matching	`35=8 150=0 39=0`	Matches DSE
PH-2	Market order during Pre-Open	Rejected (no market orders in the auction)	`35=8 150=8 39=8 58=market orders are not accepted during the Pre-Open call auction — send a limit order`	Matches DSE

ID	Case	DSE / OMS expects	Matching engine did	Result
PH-3	Limit far outside the band during Pre-Open	Rejected — circuit limits apply in every session	`35=8 150=8 39=8 58=price band: 150.00 outside +/-10.00% limit [90.00-110.00] of prev 100.00`	Matches DSE
PH-4	Uncross at the Pre-Open -> Continuous transition	Matching orders trade at the equilibrium price	`35=8 150=F 39=2 14=100`	Matches DSE
PH-5	GTC order survives a phase change	Still on the book	`no response`	Matches DSE

11. Matching

ID	Case	DSE / OMS expects	Matching engine did	Result
MT-1	Buy limit crosses the best ask — price improvement	Executes at the resting (better) price 100.00	`35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
MT-2	Price-time priority (two asks at the same price)	Earlier order fills first	`35=8 150=F 39=2 14=100 ; 35=8 150=0 39=0 ; 35=8 150=F 39=2 14=100`	Matches DSE
MT-3	Partial fill then rest (day limit)	Fills 40, rests the remaining 60	`35=8 150=0 39=0 ; 35=8 150=F 39=1 14=40`	Matches DSE

12. Rules not modelled

ID	Case	DSE / OMS expects	Matching engine did	Result
ST-1	Day orders expire at market close	Day orders are removed at close and reported	`at the end of the day (Closed) expireDayOrders removes Day orders and reports 150=C; GTC/GTD stay`	Matches DSE
ST-2	Post-close session: Market order priced at the close price	DSE requires the close price on market orders after close	`Post-Close accepts only orders at the closing price; a market order is priced at the close`	Matches DSE
ST-3	Automatic circuit breaker / volatility halt	Instrument halts on breaching limits; DSE halts per board rules	`only a static price-band check plus the manual per-symbol halt added in the admin console; no automatic trigger` not modelled	Gap
ST-4	Category-based rules (A/B/G/N/Z) and board-specific rules	Different lot sizes / bands / sessions by category and board	`Instrument has Board/ISIN/Tick only; no category, market lot, face value; band tiers are board-independent` not modelled	Gap
ST-5	Pre-trade RMS (buying power, holdings, position limit) on the FIX path	OMS-side check; the exchange simulator should also enforce	`rms.go RiskEngine.Check is called only from bench.go:352, never from the FIX acceptor` OMS performs these checks; the ME does not duplicate them	Info

ID	Case	DSE / OMS expects	Matching engine did	Result
ST-6	Order-status / execution-report replay on reconnect	Missed ERs recoverable (ResendRequest honoured)	`in-memory message store only (quickfix.NewMemoryStoreFactory) with ResetOnLogon=Y — nothing survives a reconnect` by design for a simulator	Info

13. Session messages

ID	Case	DSE / OMS expects	Matching engine did	Result
SS-1	Logon (35=A) with ResetSeqNumFlag 141=Y	Session established	`initiator sent 141=Y (ResetOnLogon=Y); session reached logged-on state`	Matches DSE
SS-2	TestRequest (35=1, 112=PING)	Heartbeat (35=0) echoing 112	`35=0`	Matches DSE
SS-3	ResendRequest (35=2, 7=1, 16=0)	Missed messages resent or gap-filled	`35=(log) ; 35=(log) ; 35=(log) ; 35=(log)`	Matches DSE
SS-4	SequenceReset (35=4, 123=N, 36=5000)	Accepted; peer's expected sequence moves	`no response`	Matches DSE
SS-5	Logout (35=5)	Logout acknowledged, session closed	`35=5`	Matches DSE
SS-6	Heartbeat (35=0) interval 30 s	Sent automatically	`handled by the QuickFIX session layer (HeartBtInt negotiated at logon)`	Matches DSE

Checklist

☑ matches DSE · ☐ needs work (Blocker = engine accepts/rejects the opposite of DSE or emits non-FIX output; Gap = feature missing or silently ignored).

Order type (40)

- ☑ `OT-01` Limit (40=2) with price
- ☑ `OT-02` Limit (40=2) without price (44), liquid book
- ☑ `OT-03` Market (40=1) without price, liquid book
- ☑ `OT-04` Market (40=1) carrying a price outside the post-close session
- ☐ `OT-05` Market (40=1) on an empty book **Info**
- ☑ `OT-06` Market-At-Best (40=Z)
- ☐ `OT-07` Stop (40=3) with StopPx (99) **Info**
- ☐ `OT-08` Stop-limit (40=4) with StopPx and price **Info**
- ☑ `OT-095` Unsupported OrdType 40=5
- ☑ `OT-099` Unsupported OrdType 40=9

Validity (59)

- ☑ `TIF-0` Day (59=0) resting limit
- ☑ `TIF-1` GTC (59=1) resting limit
- ☑ `TIF-3a` IOC (59=3) partially fillable (50 of 100)
- ☑ `TIF-3b` IOC (59=3) with no liquidity
- ☑ `TIF-4a` FOK (59=4) fully fillable
- ☑ `TIF-4b` FOK (59=4) insufficient liquidity
- ☑ `TIF-6a` GTD (59=6) without expiry date (OMS sends none)
- ☑ `TIF-6b` GTD (59=6) with ExpireTime (126) 3 s ahead — expires
- ☑ `TIF-6c` GTD (59=6) with ExpireDate (432) only
- ☐ `TIF-2` At the Opening (59=2) **Gap**
- ☐ `TIF-5` Good Till Crossing (59=5) **Gap**
- ☐ `TIF-7` At the Close (59=7) **Gap**
- ☑ `TIF-S` Session (59=S) — offered by the OMS trader UI
- ☑ `TIF-X` Missing TimeInForce

Side & quantity

- ☑ `SD-1` Buy (54=1)
- ☑ `SD-2` Sell (54=2)
- ☑ `SD-3` Missing Side (54)
- ☑ `SD-4` Side 54=5 (sell short)
- ☑ `QT-1` OrderQty = 0
- ☑ `QT-2` Fractional OrderQty (10.5)

- ☑ QT-3 Negative OrderQty
- ☑ QT-4 Quantity a multiple of 1000
- ☑ QT-5 Quantity 1500 on a lot-1000 instrument
- ☑ QT-5b Quantity 2000 on a lot-1000 instrument
- ☑ QT-6 Duplicate ClOrdID (11) resubmitted
- ☑ QT-7 Missing Symbol (55)
- ☑ QT-8 Unknown symbol

Board group (762)

- ☑ BG-01 762=PUBLIC on a MAIN-board equity
- ☐ BG-02 762=ATBPUB on a MAIN-board equity **Gap**
- ☐ BG-03 762=SPUBLIC on a MAIN-board equity **Gap**
- ☐ BG-04 762=BLOCK on a MAIN-board equity **Gap**
- ☐ BG-05 762=SBLOCK on a MAIN-board equity **Gap**
- ☐ BG-06 762=BUYIN on a MAIN-board equity **Gap**
- ☐ BG-07 762=ATBBUYIN on a MAIN-board equity **Gap**
- ☐ BG-08 762=SBUYIN on a MAIN-board equity **Gap**
- ☐ BG-09 762=DEBT on a MAIN-board equity **Gap**
- ☐ BG-10 762=ATBDEBT on a MAIN-board equity **Gap**
- ☐ BG-11 762=SDEBT on a MAIN-board equity **Gap**
- ☐ BG-12 762=ALTDBT on a MAIN-board equity **Gap**
- ☐ BG-13 762=BUYDBT on a MAIN-board equity **Gap**
- ☐ BG-14 762=YIELDDBT on a MAIN-board equity **Gap**
- ☐ BG-15 762=INDEX on a MAIN-board equity **Gap**
- ☐ BG-16 762=ATBX on a MAIN-board equity **Gap**
- ☐ BG-17 762=SCPX on a MAIN-board equity **Gap**
- ☑ BG-18 762=SPUBLIC on an SME-board equity
- ☑ BG-19 762=ATBPUB on an ATB-board equity
- ☑ BG-20 Missing 762

ExecInst (18)

- ☐ EI-1 18=Placed (n) on a new order **Partial**
- ☐ EI-2 18=Park (S) on a new order **Gap**
- ☐ EI-3 18=Block (w) on a new order **Partial**
- ☐ EI-4 18=Block Park (w S) on a new order **Gap**
- ☐ EI-5 18=q (Unpark) on a modify (35=G) **Partial**
- ☐ EI-6 18=w q (Unpark) on a modify (35=G) **Gap**

Quantity options

- ☑ QO-1 DisplayQty (1138=30) iceberg on a 100-lot (the OMS's tag)
- ☐ QO-2 MaxFloor (111=30) iceberg (engine's own tag) **Info**
- ☐ QO-3 MinQty (110) greater than OrderQty (500 > 100) **Gap**
- ☐ QO-4 MinQty (110=50) with only 30 available **Gap**

Tags & reports

- ☑ TG-1 Party group 453 (448/447/452) present
- ☑ TG-2 No party group
- ☑ TG-3 OrderRestrictions 529=N (non-standard)
- ☐ TG-4 Missing Account (1) **Info**
- ☑ TG-5 Missing TransactTime (60)
- ☑ TG-6 Missing ClOrdID (11)
- ☑ ER-1 Fill ER carries FIX-required tags (37,17,150,39,55,54,151,14,11)
- ☑ ER-2 Fill ER ExecType (150)
- ☑ ER-3 ER echoes Account (1)
- ☑ ER-4 ER echoes OrdType (40) / TimeInForce (59)
- ☑ ER-5 ER carries last fill price/qty (31/32)
- ☑ ER-6 ER carries average price (6) and cumulative qty (14)
- ☑ ER-7 ER identifies the instrument by Symbol (55) only, as the DSE does
- ☑ ER-8 Reject ER carries OrdRejReason (103) and Text (58)

Price band / tick

- ☑ All circuit-limit tier boundaries (64/64 probes correct)
- ☑ PT-1 Price on tick (100.10)
- ☑ PT-2 Price off tick (100.05)
- ☑ PT-3 Price 100.02 (sub-paisa tick)
- ☐ PT-4 Market order ignores the price band **Info**
- ☑ PT-5 Order on a halted instrument
- ☑ PT-6 Order after the halt is lifted

Cancel / modify / status

- ☑ CM-1 Cancel a resting order (35=F)
- ☑ CM-2 Cancel an unknown order
- ☑ CM-3 35=9 carries OrderID (37) — required by FIX
- ☑ CM-4 Modify price WITHOUT Symbol/762 (as the OMS sends 35=G)
- ☑ CM-5 Modify quantity 100 -> 60
- ☑ CM-6 Modify an unknown order
- ☑ CM-7 Modify to a price off tick (96.05)

- ☑ CM-8 Modify to a price outside the band (150.00)
- ☑ CM-9 Modify using the latest ClOrdID chain (replace a replaced order)
- ☑ CM-10 Replace ER echoes OrigClOrdID (41)
- ☑ CM-11 Order Status Request (35=H) on a live order
- ☑ CM-12 Order Status Request (35=H) on an unknown order
- ☐ CM-13 Modify a parked stop order **Info**

Phases

- ☑ PH-1 Limit order during Pre-Open
- ☑ PH-2 Market order during Pre-Open
- ☑ PH-3 Limit far outside the band during Pre-Open
- ☑ PH-4 Uncross at the Pre-Open -> Continuous transition
- ☑ PH-5 GTC order survives a phase change

Matching

- ☑ MT-1 Buy limit crosses the best ask — price improvement
- ☑ MT-2 Price-time priority (two asks at the same price)
- ☑ MT-3 Partial fill then rest (day limit)

Rules not modelled

- ☑ ST-1 Day orders expire at market close
- ☑ ST-2 Post-close session: Market order priced at the close price
- ☐ ST-3 Automatic circuit breaker / volatility halt **Gap**
- ☐ ST-4 Category-based rules (A/B/G/N/Z) and board-specific rules **Gap**
- ☐ ST-5 Pre-trade RMS (buying power, holdings, position limit) on the FIX path **Info**
- ☐ ST-6 Order-status / execution-report replay on reconnect **Info**

Session messages

- ☑ SS-1 Logon (35=A) with ResetSeqNumFlag 141=Y
- ☑ SS-2 TestRequest (35=1, 112=PING)
- ☑ SS-3 ResendRequest (35=2, 7=1, 16=0)
- ☑ SS-4 SequenceReset (35=4, 123=N, 36=5000)
- ☑ SS-5 Logout (35=5)
- ☑ SS-6 Heartbeat (35=0) interval 30 s

Evidence — sample messages for each deviation

SOH delimiters shown as "|". The sent string is the OMS-style message as it left the test initiator; session header fields are added by QuickFIX.

TIF-2 — At the Opening (59=2) Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=21|49=TESTBRK|50=TRADER1|52=20261017-05:56:40.895|56=DSEME|1=BO1001|11=CF00020|
38=100|40=2|44=92.00|54=1|55=T012M|59=2|60=20261017-05:56:40.895|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=189|
```

RECEIVED

```
8=FIXT.1.1|9=259|35=8|34=29|49=DSEME|52=20261017-05:56:40.895|56=TESTBRK|57=TRADER1|1=BO1001|6=0.00|
11=CF00020|14=0|17=6AD30E101C|37=2026101722|38=100|39=0|40=2|44=92.00|54=1|55=T012M|59=2|
60=20261017-05:56:40.895|150=0|151=100|453=1|448=CLIENT01|447=C|452=5|529=N|762=PUBLIC|10=148|
```

accepted but silently treated as Day (mapTIF default) — semantics not implemented

TIF-5 — Good Till Crossing (59=5) Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=22|49=TESTBRK|50=TRADER1|52=20261017-05:56:41.016|56=DSEME|1=BO1001|11=CF00021|
38=100|40=2|44=92.00|54=1|55=T012M|59=5|60=20261017-05:56:41.016|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=166|
```

RECEIVED

```
8=FIXT.1.1|9=259|35=8|34=30|49=DSEME|52=20261017-05:56:41.016|56=TESTBRK|57=TRADER1|1=BO1001|6=0.00|
11=CF00021|14=0|17=6AD30E101D|37=2026101723|38=100|39=0|40=2|44=92.00|54=1|55=T012M|59=5|
60=20261017-05:56:41.016|150=0|151=100|453=1|448=CLIENT01|447=C|452=5|529=N|762=PUBLIC|10=118|
```

accepted but silently treated as Day (mapTIF default) — semantics not implemented

TIF-7 — At the Close (59=7) Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=23|49=TESTBRK|50=TRADER1|52=20261017-05:56:41.137|56=DSEME|1=BO1001|11=CF00022|
38=100|40=2|44=92.00|54=1|55=T012M|59=7|60=20261017-05:56:41.137|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=178|
```

RECEIVED

```
8=FIXT.1.1|9=259|35=8|34=31|49=DSEME|52=20261017-05:56:41.137|56=TESTBRK|57=TRADER1|1=BO1001|6=0.00|
11=CF00022|14=0|17=6AD30E101E|37=2026101724|38=100|39=0|40=2|44=92.00|54=1|55=T012M|59=7|
60=20261017-05:56:41.137|150=0|151=100|453=1|448=CLIENT01|447=C|452=5|529=N|762=PUBLIC|10=132|
```

accepted but silently treated as Day (mapTIF default) — semantics not implemented

BG-02 — 762=ATBPUB on a MAIN-board equity Gap

SENT

```
8=FIXT.1.1|9=200|35=D|34=41|49=TESTBRK|50=TRADER1|52=20261017-05:56:43.309|56=DSEME|1=BO1001|11=CF00039|
38=100|40=2|44=95.00|54=1|55=T014M|59=0|60=20261017-05:56:43.309|453=1|448=CLIENT01|447=C|452=5|529=N|
762=ATBPUB|10=189|
```

RECEIVED

```
8=FIXT.1.1|9=259|35=8|34=49|49=DSEME|52=20261017-05:56:43.309|56=TESTBRK|57=TRADER1|1=BO1001|6=0.00|
11=CF00039|14=0|17=6AD30E102F|37=2026101733|38=100|39=0|40=2|44=95.00|54=1|55=T014M|59=0|
60=20261017-05:56:43.309|150=0|151=100|453=1|448=CLIENT01|447=C|452=5|529=N|762=ATBPUB|10=154|
```

accepted — tag 762 is never read, so board groups are not validated against the instrument

EI-2 — 18=Park (S) on a new order Gap

SENT

```
8=FIXT.1.1|9=205|35=D|34=61|49=TESTBRK|50=TRADER1|52=20261017-05:56:45.723|56=DSEME|1=BO1001|11=CF00059|18=S|
38=100|40=2|44=95.00|54=1|55=T018M|59=0|60=20261017-05:56:45.723|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=201|
```

RECEIVED

```
8=FIXT.1.1|9=259|35=8|34=69|49=DSEME|52=20261017-05:56:45.723|56=TESTBRK|57=TRADER1|1=BO1001|6=0.00|
11=CF00059|14=0|17=6AD30E1043|37=2026101753|38=100|39=0|40=2|44=95.00|54=1|55=T018M|59=0|
60=20261017-05:56:45.723|150=0|151=100|453=1|448=CLIENT01|447=C|452=5|529=N|762=PUBLIC|10=152|
```

accepted but the order rests on the book — Park is not implemented (tag 18 never read)

EI-4 — 18=Block Park (w S) on a new order Gap

SENT

```
8=FIXT.1.1|9=207|35=D|34=63|49=TESTBRK|50=TRADER1|52=20261017-05:56:45.965|56=DSEME|1=BO1001|11=CF00061|18=w
S|38=100|40=2|44=95.00|54=1|55=T020M|59=0|60=20261017-05:56:45.965|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=102|
```

RECEIVED

```
8=FIXT.1.1|9=142|35=3|34=71|49=DSEME|52=20261017-05:56:45.965|56=TESTBRK|57=TRADER1|45=63|58=Value is
incorrect (out of range) for this tag|371=18|372=D|373=5|10=088|
```

rejected — 'w S' is not in the engine's tag-18 dictionary enum, so the OMS's DSE-specific instruction never reaches the engine

EI-6 — 18=w q (Unpark) on a modify (35=G) Gap

SENT

```
8=FIXT.1.1|9=198|35=G|34=67|49=TESTBRK|50=TRADER1|52=20261017-05:56:46.448|56=DSEME|1=BO1001|11=CF00065|18=w
q|38=100|40=2|41=CF00064|44=95.00|54=1|59=0|60=20261017-05:56:46.448|453=1|448=CLIENT01|447=C|452=5|529=N|
10=065|
```

RECEIVED

```
8=FIXT.1.1|9=142|35=3|34=75|49=DSEME|52=20261017-05:56:46.448|56=TESTBRK|57=TRADER1|45=67|58=Value is
incorrect (out of range) for this tag|371=18|372=G|373=5|10=096|
```

rejected — Unpark ('w q') cannot be sent through the dictionary-validated session

QO-3 — MinQty (110) greater than OrderQty (500 > 100) Gap

SENT

```
8=FIXT.1.1|9=208|35=D|34=70|49=TESTBRK|50=TRADER1|52=20261017-05:56:46.810|56=DSEME|1=BO1001|11=CF00068|
38=100|40=2|44=90.00|54=1|55=T024M|59=0|60=20261017-05:56:46.810|110=500|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=043|
```

RECEIVED

```
8=FIXT.1.1|9=259|35=8|34=78|49=DSEME|52=20261017-05:56:46.810|56=TESTBRK|57=TRADER1|1=BO1001|6=0.00|
11=CF00068|14=0|17=6AD30E104A|37=2026101759|38=100|39=0|40=2|44=90.00|54=1|55=T024M|59=0|
60=20261017-05:56:46.810|150=0|151=100|453=1|448=CLIENT01|447=C|452=5|529=N|762=PUBLIC|10=160|
```

accepted — MinQty (110) is ignored

QO-4 — MinQty (110=50) with only 30 available Gap

SENT

```
8=FIXT.1.1|9=208|35=D|34=71|49=TESTBRK|50=TRADER1|52=20261017-05:56:46.931|56=DSEME|1=BO1001|11=CF00069|
38=100|40=2|44=100.00|54=1|55=T025M|59=0|60=20261017-05:56:46.931|110=50|453=1|448=CLIENT01|447=C|452=5|529=N|
762=PUBLIC|10=046|
```

RECEIVED

```
8=FIXT.1.1|9=260|35=8|34=79|49=DSEME|52=20261017-05:56:46.931|56=TESTBRK|57=TRADER1|1=BO1001|6=0.00|
11=CF00069|14=0|17=6AD30E104B|37=2026101761|38=100|39=0|40=2|44=100.00|54=1|55=T025M|59=0|
60=20261017-05:56:46.931|150=0|151=100|453=1|448=CLIENT01|447=C|452=5|529=N|762=PUBLIC|10=197|
8=FIXT.1.1|9=327|35=8|34=80|49=DSEME|52=20261017-05:56:46.931|56=TESTBRK|57=TRADER1|1=BO1001|6=100.00|
11=CF00069|14=30|17=6AD30E104C|31=100.00|32=30|37=2026101761|38=100|39=1|40=2|44=100.00|54=1|55=T025M|59=0|
60=20261017-05:56:46.931|64=20261019|75=20261017|150=F|151=70|381=3000.00|453=1|448=CLIENT01|447=C|452=5|
529=N|762=PUBLIC|880=5|1057=Y|10=038|
```

filled below the minimum quantity — MinQty (110) is ignored

Recommended next work

#	Priority	Change	Where
1	Gap High	**Validity At-Open / GTX / At-Close (59=2/5/7)** Rest 59=2 for the opening auction only, 59=7 for the close; 59=5 expires at the next uncross. — TIF-2, TIF-5, TIF-7	fixacceptor.go mapTIF, engine.go
2	Gap High	**ExecInst Park / Unpark / Block (18)** Add 'w S' and 'w q' to the dictionary; hold parked orders off the book until unparked. — EI-1 … EI-6	spec/FIX50SP2.xml, fixacceptor.go, engine.go

#	Priority	Change	Where
3	Gap Medium	**MinQty (110)** Reject MinQty > OrderQty; only trade when at least MinQty can fill at once. — QO-3, QO-4	fixacceptor.go, engine.go match
4	Gap Medium	**Validate tag 762 against the instrument's board** Map board groups to boards; reject index groups for orders. — BG-02 … BG-17	fixacceptor.go validateNew
5	Gap Low	**Automatic circuit breaker; category and board rules** Halt on band breaches per DSE rules; per-category lots, bands and sessions. — ST-3, ST-4	engine.go, instruments

Out of scope for this report: OMS-side defects listed in the OMS reference (GTD sent without expiry, modify without Symbol/762, Market-At-Best blocked by the OMS permission check) — they are on the sending side. Items DSE has never been observed to accept or reject (block, buy-in, iceberg, GTD, cancel, modify, status) are judged against the FIX standard and the OMS reference only and should be re-confirmed against DSE.